

Fast Signature Based Generative Models for Financial Time Series

Gordon Lee
BNY

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Abstract

This paper introduces and compares two computationally efficient, path-signature-based methods for generative modeling of financial time series. We begin with a fully non-parametric, path-wise bootstrap model that resamples historical segments. We then introduce a fully parametric Signature-SDE model, where both drift and diffusion are functions of the path signature, estimated using Kernel Ridge Regression. We evaluate the models on their ability to learn complex, mean-reverting stochastic processes, using a linear signature-based Maximum Mean Discrepancy (MMD) metric to compare the generated path distributions. Our results show that this produces a parametric model, producing high-fidelity synthetic paths at low computational cost.

1 Introduction

The accurate modeling of financial time series is a cornerstone of quantitative finance. A key challenge is the development of generative models that can produce realistic synthetic data, capturing stylized facts like volatility clustering and heavy tails. While classical models like GARCH [2] provide foundational insights, and modern methods like Neural SDEs [3] offer great flexibility, there exists a need for methods that are both powerful and computationally tractable.

This paper explores the path signature, a concept from rough path theory [4], as a foundation for such models. The signature provides a rich, non-parametric summary of a path’s history, making it an ideal feature set. While its use in discriminative tasks is growing [6], its application in a generative context is less explored. Many recent generative models that use signatures have started to lean on neural networks, inheriting their high computational cost. This work aims to provide an alternative, showing how to build fast and effective generative models without resorting to complex neural architectures.

We propose and implement two distinct, computationally efficient generative models that leverage the path signature:

1. **A Path-wise Bootstrap Model:** A simple, fully non-parametric k-NN resampling method that serves as a robust baseline.
2. **Kernel-Ridge Estimated SDE Model:** A direct estimate of the drift and volatility function of an SDE using signatures as input

We provide a rigorous comparison of these methods, demonstrating the journey from a simple baseline, through the challenges of a highly flexible model, to a stable and effective hybrid solution. We evaluate performance using a linear signature-based Maximum Mean Discrepancy (MMD) test.

2 Background: The Path Signature

The signature of a path is a fundamental concept from Terry Lyons’ theory of rough paths [4]. It provides a canonical way to describe a continuous path in Euclidean space, effectively serving as a non-parametric feature set that captures the path’s essential geometric properties.

2.1 Mathematical Definition

Let $X : [0, T] \rightarrow \mathbb{R}^d$ be a continuous path of finite variation. The signature of X , denoted $\mathcal{S}(X)_{0,T}$, is an infinite sequence of iterated integrals, living in the tensor algebra space over $\mathbb{R}^d$. The k -th term in the sequence is a tensor of order k whose components are given by:

$$\mathcal{S}(X)_{0,T}^{(k)} = \int_{0 < t_1 < \dots < t_k < T} dX_{t_k}^{i_k} \dots dX_{t_1}^{i_1}$$

where $i_j \in \{1, \dots, d\}$. In practice, the signature is truncated at a certain level m . For our application, we model a one-dimensional price path S_t by creating a two-dimensional path $X_t = (t, S_t)$, which allows the signature to capture time-dependent effects.

2.2 Key Properties and Theoretical Power

The signature is more than just a collection of features; it possesses deep mathematical properties that make it uniquely suited for path analysis.

- **Algebraic Properties:** The signature respects the concatenation of paths via Chen’s identity, $\mathcal{S}(X_{0,T}) = \mathcal{S}(X_{0,u}) \otimes \mathcal{S}(X_{u,T})$, where $\otimes$ is the tensor product. This provides a principled way to update the representation of a path as new data arrives.
- **Uniqueness and Faithfulness:** For a broad class of paths, the signature uniquely determines the path up to tree-like deviations. This means the signature is a faithful and unambiguous representation of the path’s history.
- **Relation to Expected Values:** A crucial property is that the expected signature, $\mathbb{E}[\mathcal{S}(X)]$, uniquely determines the law (the probability distribution) of the stochastic process generating the path X [7]. This is the theoretical foundation for our evaluation metric: if the expected signatures of two path distributions are close, then the distributions themselves are close.

This final point is what motivates the use of the linear signature MMD as a powerful tool for comparing generative models of time series.

2.3 Model 1: Path-wise Bootstrap

This model is a fully non-parametric baseline that approximates the conditional distribution by resampling from an empirical library. The procedure is outlined in Algorithm 1.

2.3.1 Library Creation

The model constructs an empirical library of historical cause-and-effect pairs, $\mathcal{D} = \{(\mathbf{s}_i, \mathbf{p}_i)\}$. Here, $\mathbf{s}_i = \mathcal{S}(X_{[t_i-L, t_i]})$ is the signature of a past lookback window, representing the "cause". The "effect" is the corresponding future path segment $\mathbf{p}_i = X_{[t_i+1, t_i+1+F]} - X_{t_i}$, stored as a relative increment to ensure continuity.

2.3.2 Path Generation

To generate a future segment, we are effectively drawing from an empirical approximation of the true conditional distribution:

1. Given the current generated path X_{gen} , compute its lookback signature $\mathbf{s}_{gen}$.
2. Find the k -Nearest Neighbors (k -NN) of $\mathbf{s}_{gen}$ in the library $\mathcal{D}$. Let this neighborhood be $\mathcal{N}_k(\mathbf{s}_{gen})$.
3. The conditional distribution is approximated by the empirical distribution of the future segments in this neighborhood. We draw a sample by picking one of these segments uniformly at random: $\mathbf{p}_{sample} \sim \text{Uniform}(\{\mathbf{p}_j | (\mathbf{s}_j, \mathbf{p}_j) \in \mathcal{N}_k(\mathbf{s}_{gen})\})$.
4. The generated path is extended by appending this sampled segment.

Algorithm 1 Path-wise Signature-Bootstrap

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1: Library Creation Phase
2: Input: Historical paths  $\mathcal{R}$ , lookback  $L$ , forward window  $F$ , signature level  $m$ .
3:  $\mathcal{D} \leftarrow \emptyset$ .
4: for  $R \in \mathcal{R}$  do
5:   for  $t = L$  to  $|R| - F$  do
6:      $\mathbf{s}_t \leftarrow \mathcal{S}(R[t - L : t])$ .
7:      $\mathbf{p}_t \leftarrow R[t + 1 : t + 1 + F] - R[t]$ .
8:     Add  $(\mathbf{s}_t, \mathbf{p}_t)$  to library  $\mathcal{D}$ .
9:   end for
10: end for
11: Output: Historical library  $\mathcal{D}$ .

12: Generation Phase
13: Input: Library  $\mathcal{D}$ , seed path  $R_{seed}$ , number of steps  $N$ , neighbors  $k$ .
14:  $R_{gen} \leftarrow R_{seed}$ .
15: while  $|R_{gen}| < N$  do
16:    $\mathbf{s}_{gen} \leftarrow \mathcal{S}(R_{gen}[\text{end} - L : \text{end}])$ .
17:   Find set  $\mathcal{N}_k$  of  $k$  pairs  $(\mathbf{s}_i, \mathbf{p}_i) \in \mathcal{D}$  where  $\mathbf{s}_i$  is closest to  $\mathbf{s}_{gen}$ .
18:   Randomly sample one pair  $(\mathbf{s}_j, \mathbf{p}_j)$  from  $\mathcal{N}_k$ .
19:   Append the segment  $R_{gen}[\text{end}] + \mathbf{p}_j$  to  $R_{gen}$ .
20: end while
21: Output: Generated path  $R_{gen}$ .
```

Sketch Proof of Correctness: Path-wise Signature-Bootstrap

The goal is to prove that the distribution of paths generated by the algorithm converges to the true data-generating distribution as the historical data library $\mathcal{D}$ grows infinitely large. This is a proof of **statistical consistency**.

We begin by stating the necessary assumptions.

Assumptions

Assumption A1. Process Characteristics. The underlying data-generating process, $(X_t)_{t \geq 0}$, is assumed to be *ergodic* and *stationary*. This ensures that the time-shifted pairs $(\mathbf{s}_t, \mathbf{p}_t)$ from a single long realization can be treated as i.i.d. samples from a stationary joint distribution $\mathbb{P}$ on the space of signature-increment pairs.

Assumption A2. Continuity of Conditional Expectation. The mapping from a past signature to the conditional expectation of the future path is continuous. More formally, let $E[\cdot|\mathbf{s}]$ be the conditional expectation operator. We assume that the map $\mathbf{s} \mapsto E[f(\mathbf{p})|\mathbf{s}]$ is continuous for any bounded, continuous function f on the space of paths.

Assumption A3. Data Density. The support of the marginal distribution of the signatures $\mathbf{s}$ is the entire reachable signature space. This ensures that for any newly generated signature $\mathbf{s}_{gen}$, we can always find historical neighbors in the library.

Lemma 1 (Consistency of k-NN Resampling). Let $(\mathbf{s}_i, \mathbf{p}_i)_{i=1}^{|\mathcal{D}|}$ be samples drawn from the stationary distribution $\mathbb{P}$ as per Assumption A1. Let $\mathbf{s}_{gen}$ be a point in the signature space and $\mathcal{N}_k(\mathbf{s}_{gen})$ be the set of its k nearest neighbors in $\{\mathbf{s}_i\}$. Let $\hat{\mathbb{P}}_k(\cdot|\mathbf{s}_{gen})$ be the empirical measure of the corresponding increments $\{\mathbf{p}_j | (\mathbf{s}_j, \mathbf{p}_j) \in \mathcal{N}_k(\mathbf{s}_{gen})\}$.

Then, under Assumption A2, as $|\mathcal{D}| \rightarrow \infty$ and k is chosen such that $k \rightarrow \infty$ and $k/|\mathcal{D}| \rightarrow 0$, the measure $\hat{\mathbb{P}}_k$ converges weakly to the true conditional measure $\mathbb{P}(\cdot|\mathbf{s}_{gen})$.

Proof. This result is a direct application of the universal consistency of k-NN estimators, as established in the seminal work of Stone (1977). The convergence of the neighbors' covariates to

$\mathbf{s}_{gen}$ and the continuity from Assumption A2 ensure that the empirical measure of the responses converges to the true conditional measure. $\square$

Lemma 2 (Signature as a Sufficient Statistic). *The signature of a path provides a unique and continuous representation of its history. The conditional law of the future path segment, given the past, is a continuous function of the signature of the past path.*

Proof. This lemma is a consequence of foundational results in Rough Path Theory. Chen’s theorem establishes the injectivity of the signature map, while Lyons’ Universal Limit Theorem shows that continuous functionals on path space (like the conditional expectation of the future path) can be continuously approximated by linear functions of the signature. This validates Assumption A2. $\square$

Theorem 3 (Consistency of the Algorithm). *Let the data-generating process $(X_t)_{t \geq 0}$ satisfy assumptions A1, A2, and A3. Let R_{gen} be a path generated by the Path-wise Signature-Bootstrap algorithm. As the library size $|\mathcal{D}| \rightarrow \infty$, the finite-dimensional distributions of the generated process (R_{gen}) converge to those of the true process (X_t) .*

Proof. The proof is by induction on the number of generated segments.

1. **Base Case:** The generation begins from a real seed path from the true process.
2. **Inductive Step:** Assume the generated path is a valid realization up to the current step. The next segment is generated based on the lookback signature $\mathbf{s}_{gen}$.
3. By Lemma 2, this signature is a sufficient statistic for the path’s history.
4. By Lemma 1, the k-NN resampling step consistently estimates the true conditional distribution $\mathbb{P}(\cdot | \mathbf{s}_{gen})$. In the limit, the algorithm draws a sample from the true distribution.

By induction, every segment is drawn from the correct conditional law. Therefore, the joint distribution of the generated path converges to the joint distribution of the true process. $\square$

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2.4 Model 2: Kernel Ridge Regression in Signature Space

Our second model provides a non-parametric alternative, framing the problem within the well-established Kernel Ridge Regression (KRR) framework. Instead of explicitly parameterizing the drift and volatility functions, this model directly estimates their conditional expectation as a linear function in the feature space induced by the signature kernel. This approach is computationally efficient during generation and benefits from the rich, non-linear feature set provided by the path signature.

2.4.1 Model Formulation

The model operates on the log-price process, $X_t = \log S_t$. The goal is to model the drift and volatility of this process as functions of the signature of its recent history.

Training Phase: First, we transform the historical time series into a supervised learning dataset. For each path in our training data, we create a rolling window of length L . The features are the path signatures of these windows, and the targets are the empirically estimated drift and log-volatility.

Let $\mathcal{X} = \{X^{(1)}, \dots, X^{(N)}\}$ be a set of N historical path segments (windows). We form the signature

matrix $\mathbf{S}$ by stacking their corresponding truncated signatures:

$$\mathbf{S} = \begin{bmatrix} (\mathcal{S}(X^{(1)})_m)^\top \\ (\mathcal{S}(X^{(2)})_m)^\top \\ \vdots \\ (\mathcal{S}(X^{(N)})_m)^\top \end{bmatrix} \in \mathbb{R}^{N \times d(m)},$$

where $d(m)$ is the dimension of the signature truncated at level m . For each path segment $X^{(i)}$, we define two corresponding targets:

$$y_\mu^{(i)} = \frac{\log S_{t_{i+1}} - \log S_{t_i}}{\Delta t}, \quad y_{\log \sigma}^{(i)} = \log \left(\frac{\text{std}(\{\Delta \log S\} \text{ in window } i)}{\sqrt{\Delta t}} \right).$$

We use a linear kernel on the signature features, which defines the Gram matrix as $K = \mathbf{S}\mathbf{S}^\top \in \mathbb{R}^{N \times N}$. We then solve two separate ridge regression problems to find the model weights $(\alpha_\mu, \alpha_{\log \sigma})$:

$$\alpha_\mu = (K + \lambda I_N)^{-1} \mathbf{y}_\mu, \quad (1)$$

$$\alpha_{\log \sigma} = (K + \lambda I_N)^{-1} \mathbf{y}_{\log \sigma}. \quad (2)$$

Prediction Phase: Given a new window with signature $\mathbf{s}_{\text{new}}$, the predicted drift and log-volatility are computed via the kernel trick:

$$\hat{\mu} = (\mathbf{S} \mathbf{s}_{\text{new}})^\top \alpha_\mu, \quad \widehat{\log \sigma} = (\mathbf{S} \mathbf{s}_{\text{new}})^\top \alpha_{\log \sigma}.$$

Algorithm 2 Kernel Ridge Signature Model

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1: Training Phase
2: Input: Historical paths  $\mathcal{X}$ , lookback  $L$ , signature level  $m$ , regularization  $\lambda$ .
3: Initialize empty lists  $S_{list}$ ,  $Y_\mu$ ,  $Y_{\log \sigma}$ .
4: for each path  $X$  in  $\mathcal{X}$  do
5:   for each time step  $t$  from  $L$  to end do
6:      $W \leftarrow X_{[t-L, t]}$  ▷ Extract lookback window
7:      $\mathbf{s} \leftarrow \mathcal{S}(W)_m$  ▷ Compute signature
8:      $S_{list}.\text{append}(\mathbf{s})$ 
9:     Calculate empirical drift  $y_\mu$  and log-volatility  $y_{\log \sigma}$  from  $W$ .
10:     $Y_\mu.\text{append}(y_\mu)$ ;  $Y_{\log \sigma}.\text{append}(y_{\log \sigma})$ 
11:   end for
12: end for
13:  $\mathbf{S} \leftarrow \text{stack}(S_{list})$ ;  $\mathbf{y}_\mu \leftarrow \text{stack}(Y_\mu)$ ;  $\mathbf{y}_{\log \sigma} \leftarrow \text{stack}(Y_{\log \sigma})$ 
14:  $K \leftarrow \mathbf{S}\mathbf{S}^\top$  ▷ Form linear kernel matrix
15:  $\alpha_\mu \leftarrow (K + \lambda I)^{-1} \mathbf{y}_\mu$  ▷ Solve for drift weights
16:  $\alpha_{\log \sigma} \leftarrow (K + \lambda I)^{-1} \mathbf{y}_{\log \sigma}$  ▷ Solve for volatility weights
17: Output: Model parameters  $(\alpha_\mu, \alpha_{\log \sigma})$  and training signatures  $\mathbf{S}$ .

18: Generation Phase
19: Input: Trained model  $(\alpha_\mu, \alpha_{\log \sigma}, \mathbf{S})$ , seed path  $X_{seed}$ , steps to generate  $N_{gen}$ .
20: Initialize generated path  $X_{gen} \leftarrow X_{seed}$ .
21: for  $i$  from  $L$  to  $L + N_{gen}$  do
22:    $W_{current} \leftarrow X_{gen}[i - L : i]$  ▷ Get current window
23:    $\mathbf{s}_{new} \leftarrow \mathcal{S}(W_{current})_m$  ▷ Compute new signature
24:    $\mathbf{k} \leftarrow \mathbf{S}\mathbf{s}_{new}$  ▷ Compute kernel vector
25:    $\hat{\mu} \leftarrow \mathbf{k}^\top \alpha_\mu$ ;  $\widehat{\log \sigma} \leftarrow \mathbf{k}^\top \alpha_{\log \sigma}$  ▷ Predict
26:    $\Delta W_i \sim \mathcal{N}(0, \Delta t)$  ▷ Sample random shock
27:    $X_{gen}[i + 1] \leftarrow X_{gen}[i] + \hat{\mu}\Delta t + \exp(\widehat{\log \sigma})\Delta W_i$  ▷ Generate next point
28: end for
29: Output: Synthetic path  $X_{gen}$ .
```

2.4.2 Path Generation

Path generation is an autoregressive process. Starting from an initial seed path of length $L + 1$, we recursively generate the next point in the log-price path using the Euler-Maruyama scheme with our predicted parameters:

$$\log S_{t_{i+1}} = \log S_{t_i} + \hat{\mu} \Delta t + \exp(\widehat{\log \sigma}) \Delta W_i,$$

where $\Delta W_i \sim \mathcal{N}(0, \Delta t)$ are independent increments of a Wiener process. This procedure, detailed in Algorithm 2, yields synthetic trajectories whose conditional first and second-order statistics match the training data by construction.

2.5 Discussion and Extensions

- **Computational cost:** Training incurs an $\mathcal{O}(N^3)$ solve for α , but prediction is $\mathcal{O}(N d(m))$ per step.
- **Scalability:** One can approximate K via *Nyström* or *random signature features* to reduce runtime.
- **Distributional matching:** An MMD penalty in signature-kernel space can further align higher-order path statistics.

Algorithm 3 Bootstrap-Regularized SDE

- 1: **Training Phase**
 - 2: **Input:** Historical paths $\mathcal{X}$, lookback L , signature level m , regularization λ , bootstrap regularizer strength γ .
 - 3: Create historical dataset $\mathcal{D}_{hist} = \{(S(X_{[t-L, t]}), \Delta X_t)\}_{X \in \mathcal{X}}$.
 - 4: Create bootstrap model (Model 1) from $\mathcal{X}$.
 - 5: **function** COMPOSITELOSS($(\theta_\mu, \theta_\sigma)$)
 - 6: $L_{NLL} \leftarrow \text{NegativeLogLikelihood}(\theta_\mu, \theta_\sigma, \mathcal{D}_{hist}, \lambda)$.
 - 7: Generate SDE paths X_{SDE} using $(\theta_\mu, \theta_\sigma)$.
 - 8: Generate Bootstrap paths X_{Boot} .
 - 9: $L_{MMD} \leftarrow \text{MMD}^2(\text{law}(X_{SDE}), \text{law}(X_{Boot}))$.
 - 10: **return** $L_{NLL} + \gamma \cdot L_{MMD}$.
 - 11: **end function**
 - 12: $(\theta_\mu^*, \theta_\sigma^*) \leftarrow \arg \min_{\theta_\mu, \theta_\sigma} \text{CompositeLoss}(\theta_\mu, \theta_\sigma)$.
 - 13: **Output:** Optimal parameters $\theta_\mu^*, \theta_\sigma^*$.
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3 Experimental Setup

We evaluate the models on their ability to learn the dynamics of a known stochastic process.

3.1 Data Generation

We simulate 50 ground-truth paths from a mean-reverting SDE with time-dependent volatility: $dy_t = (\mu t - \theta y_t)dt + (2\sigma t/T)dW_t$. Each path has a length of 512 steps.

3.2 Evaluation Metric: Linear Signature MMD

We evaluate the quality of the generated paths using the linear signature MMD, which computes the squared Euclidean distance between the mean signature vectors of two empirical path sets.

4 Results and Analysis

We trained all three models and generated 50 new paths from each.

4.1 Model Performance

The Path-wise Bootstrap (Model 1) and the Bootstrap-Regularized SDE (Model 3) both produced excellent, stable results. In contrast, the standard Fully Parametric SDE (Model 2) produced unstable and explosive paths.

Table 1 summarizes the key performance metrics for the two successful models. Both achieve excellent MMD scores, indicating high fidelity. The bootstrap-regularized SDE achieves a slightly lower MMD, suggesting it captures the underlying process with the highest fidelity while remaining stable.

Metric	Path-wise Bootstrap	Bootstrap-Reg. SDE
Training Time (CPU)	Negligible ($\approx 0.5s$)	Slow (≈ 5 min)
Linear Sig-MMD ²	0.0075	0.0004
Generates Novel Paths?	No	Yes

Table 1: Comparison of the final, stable generative models.

5 Discussion

The iterative development process highlighted a crucial lesson in generative modeling: model stability is paramount. The fully parametric SDE model, despite its theoretical power, consistently failed by learning unstable parameters. The key to success was the novel bootstrap-regularization technique, which effectively constrains the optimization to a region of stable parameters. This method acts as a powerful safeguard, preventing the SDE from diverging while allowing it to learn the subtle dynamics present in the data.

6 Conclusion

In this paper, we presented and compared three fast, signature-based generative models. We showed that a fully parametric approach can be unstable, and that a more robust solution is achieved by using the distribution of a simpler, non-parametric model as a regularizer during training. Our final Bootstrap-Regularized SDE is stable, computationally tractable on CPUs, and produces paths that are statistically very close to the ground truth, as verified by the powerful linear signature MMD metric. Our work confirms that the path signature provides a practical foundation for building the next generation of generative financial models.

References

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